Optical Tweezers:
Measuring the Piconewton Forces
PHYS5381 Final Presentation

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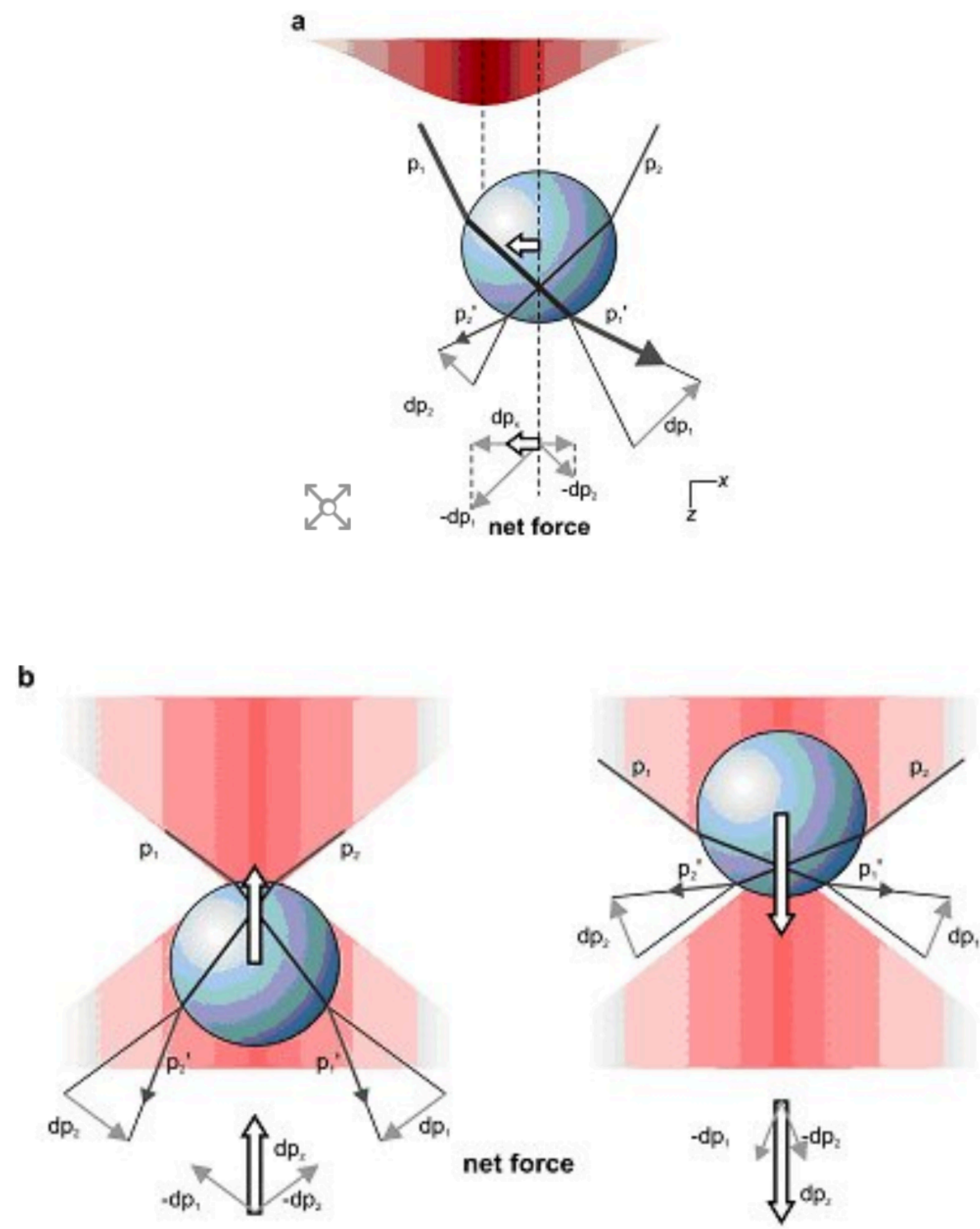
Motivation

History, Mechanisms, Application.

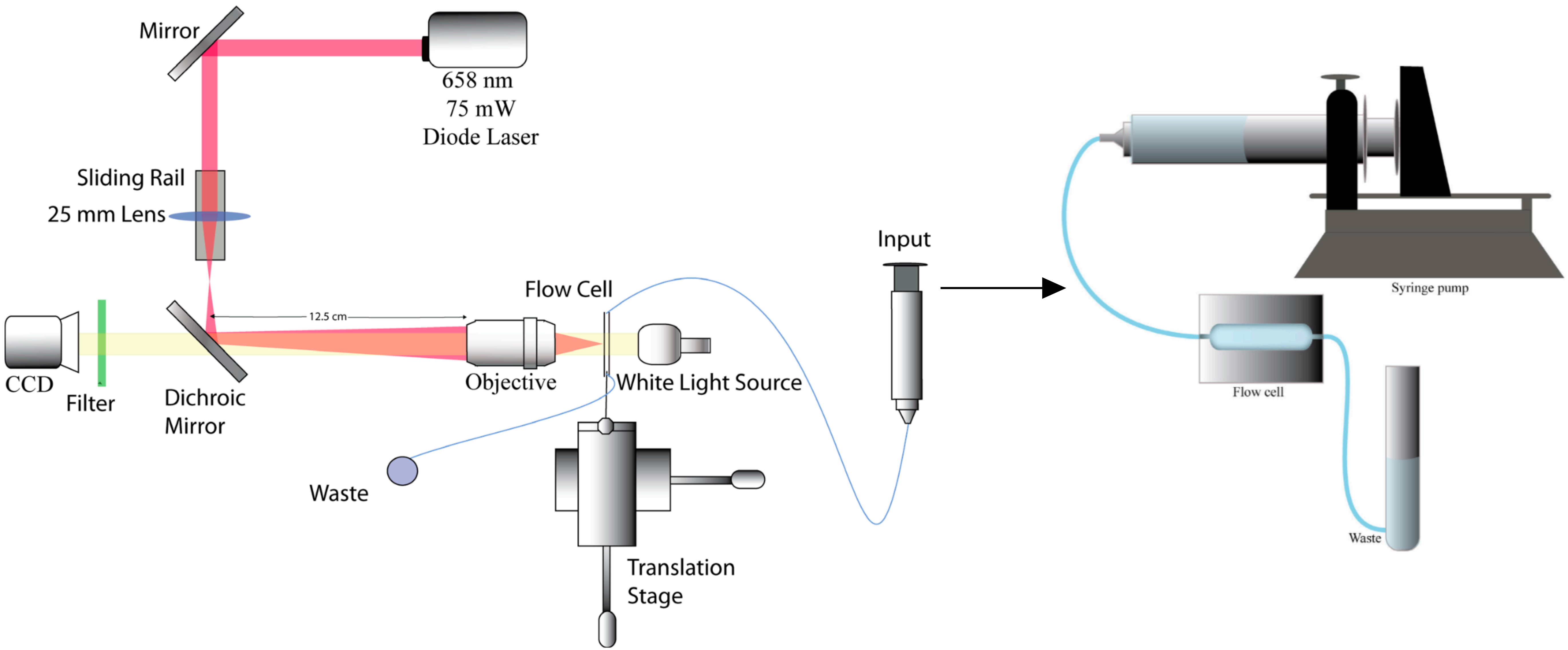
- First demonstrated by Arthur Ashkin in 1986
- Essential in biophysics Research.
  - Elasticity of DNA
  - Stepping of molecular motors
- Induced force near the focal point due to intensity gradient.
- Restores for small displacement from the equilibrium position.



https://physicsworld.com/a/optical-tweezers-where-physics-meets-biology/

Material and Methods

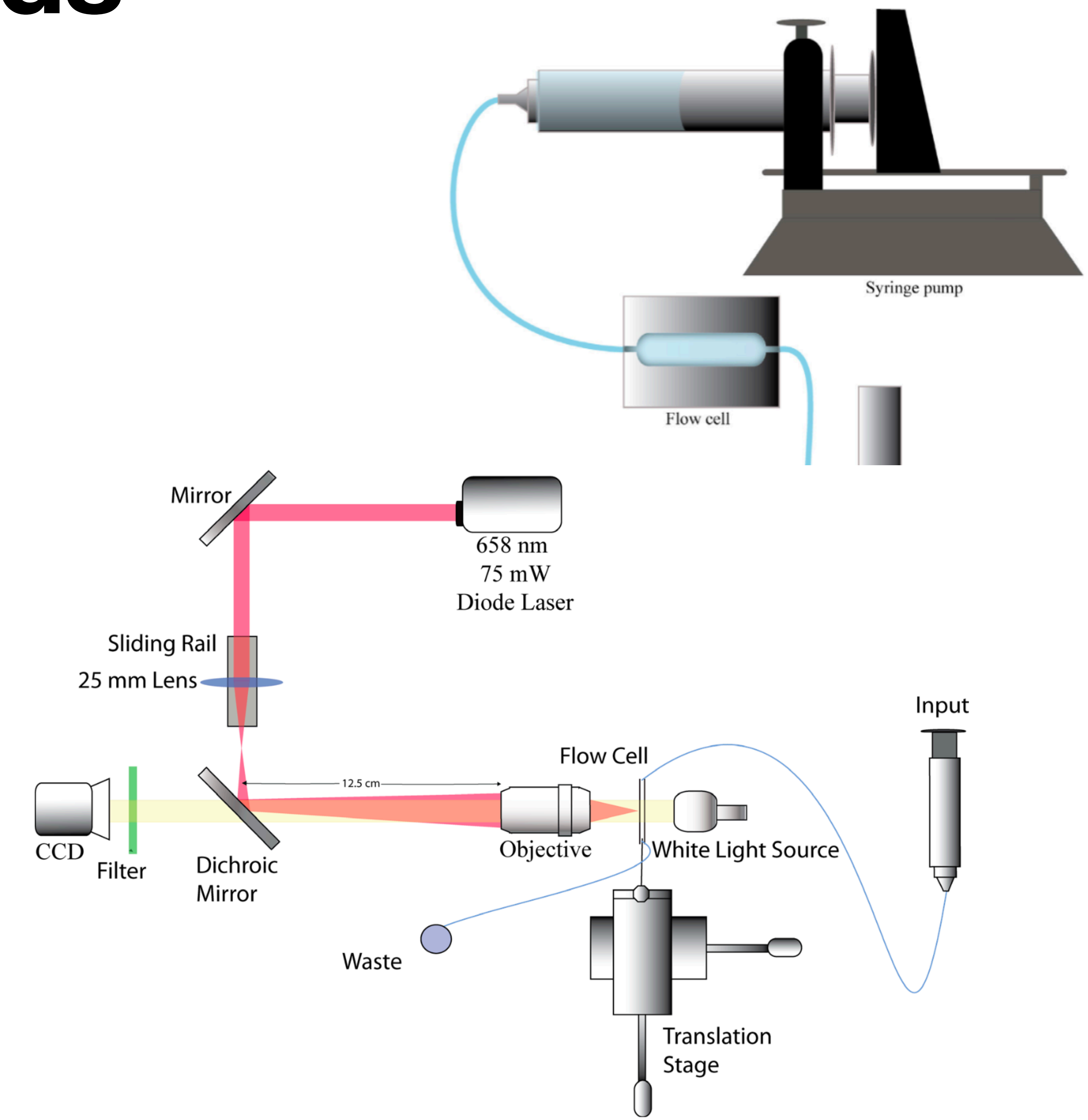
The experimental setup



Material and Methods

The experimental setup

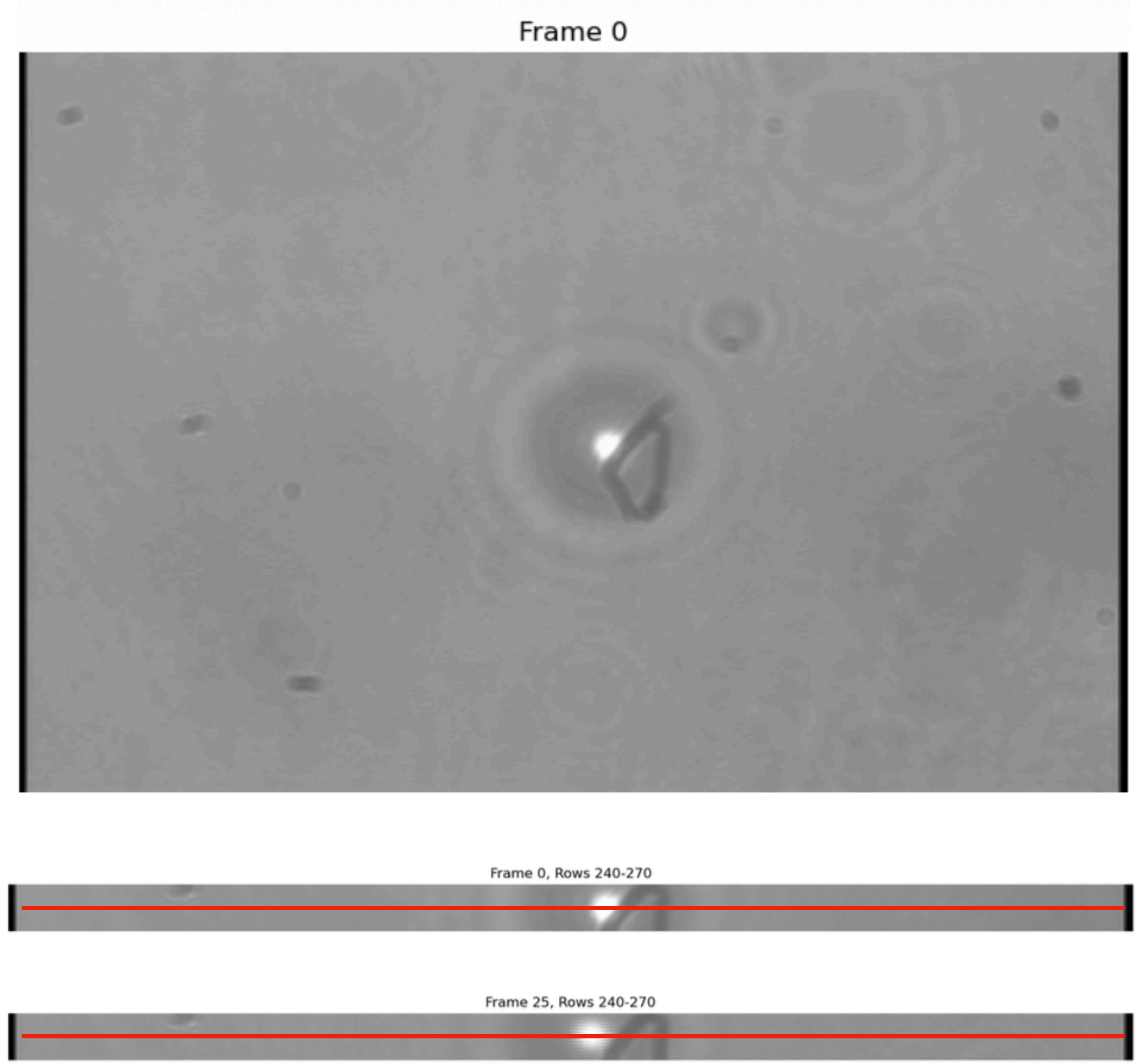
- Bead: trapped particles
  - Diameter of 4.16 ± 0.21 μm
  - Index of refraction n = 1.50
  - Deionized water n = 1.33



Material and Methods

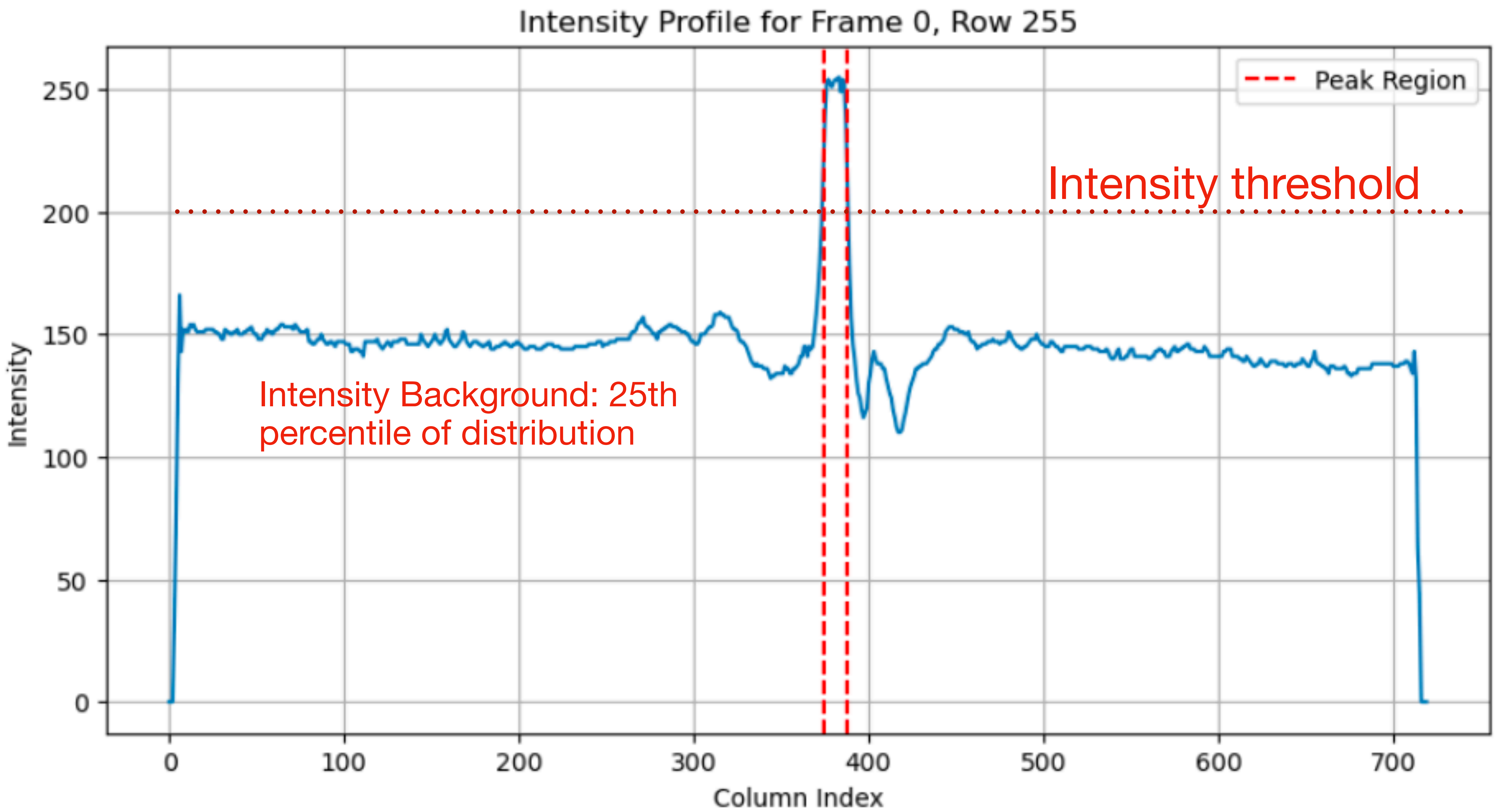
Data Acquisition

- Measurements were taken at
    - 25, 50, 75, 120, 125, 130, 135, 145 μL/min.
- Python with the OpenCV, NumPy, and SciPy libraries.
- A region of interest (ROI): rows 240-270 was selected, as it trapped bead and optical trap center



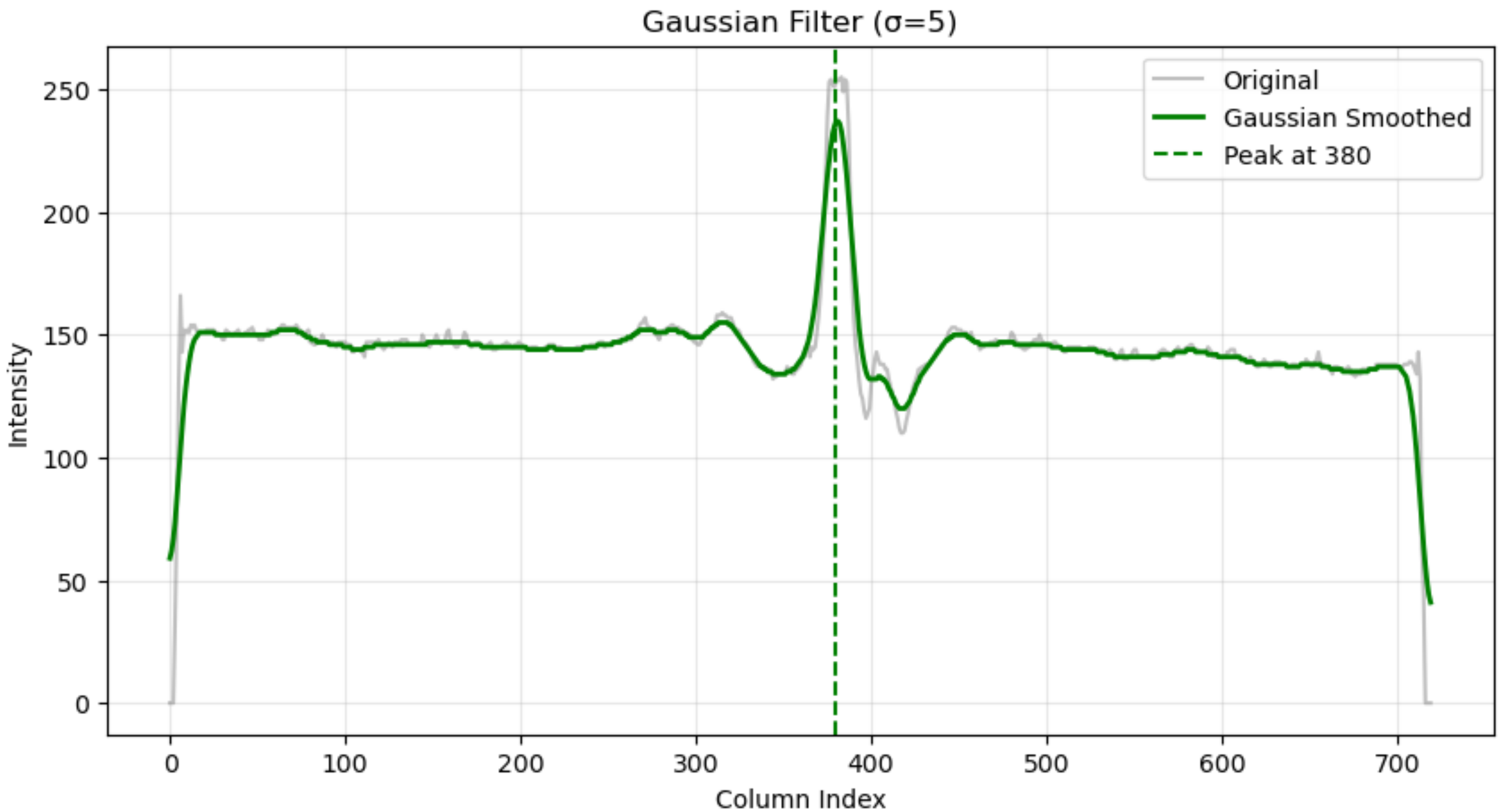
Material and Methods

Data Acquisition



Results

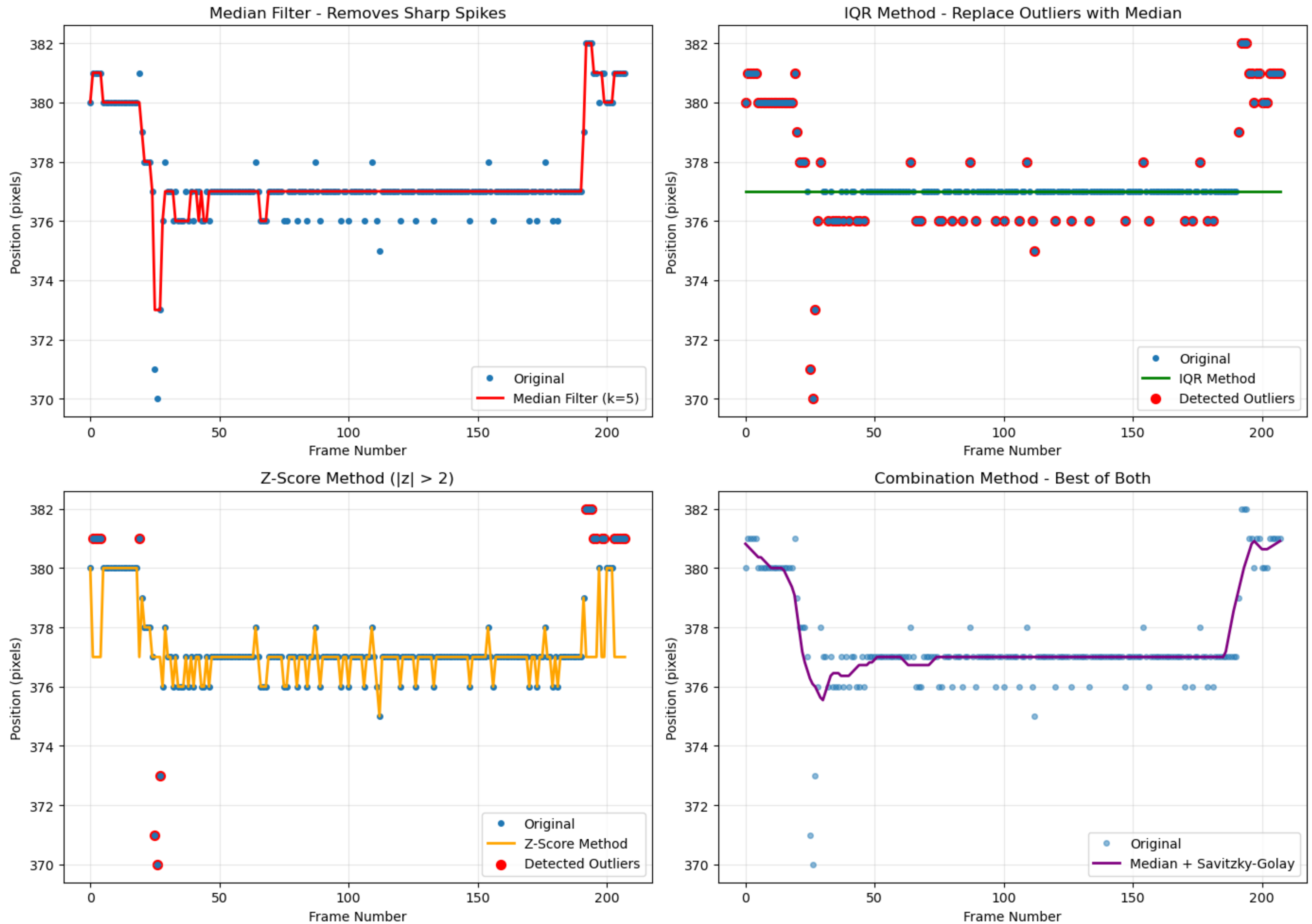
Data analysis



Results

Data analysis

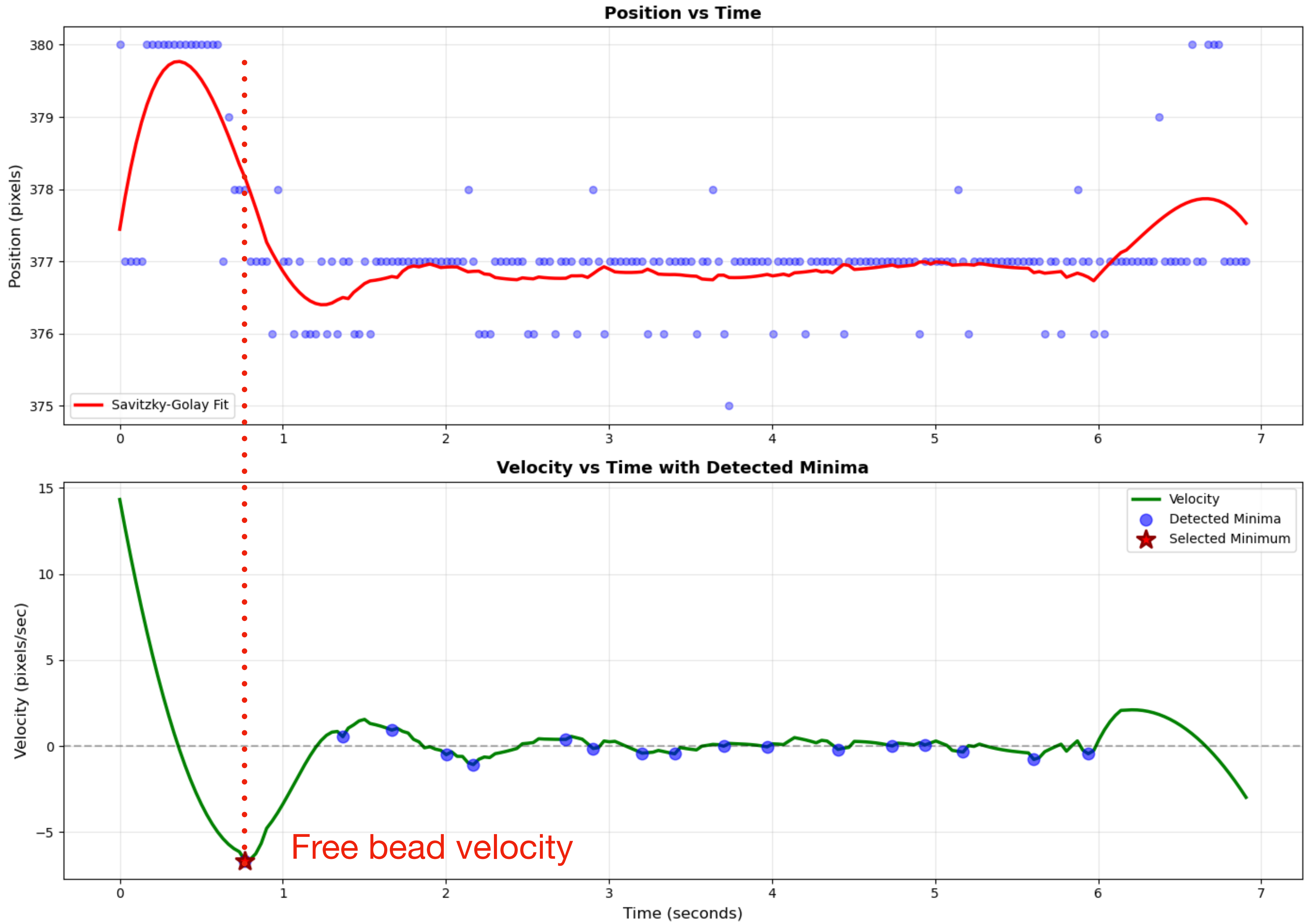
- Median Filter
- IQR Method
- Z-Score
- Combined



Results

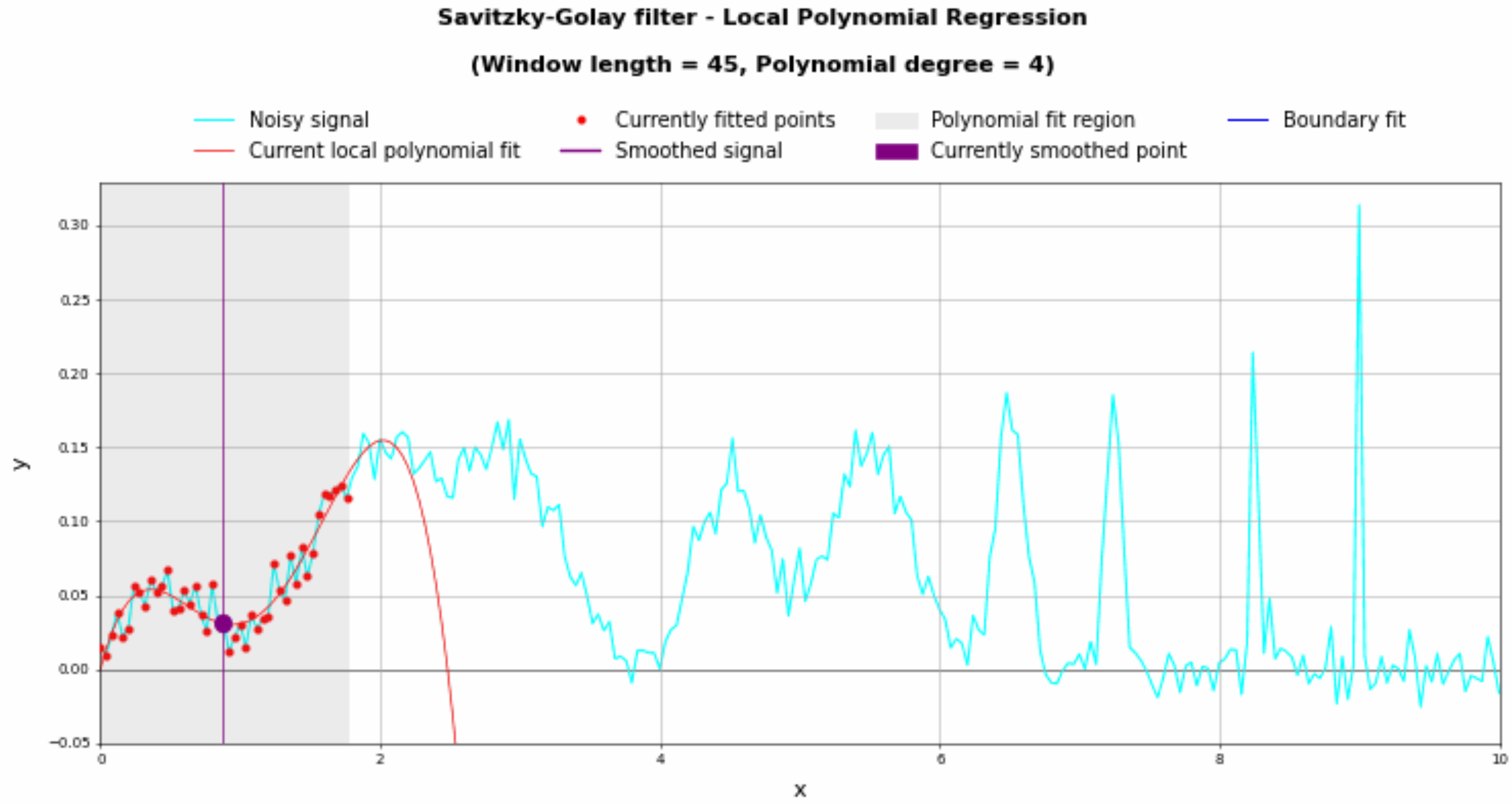
Data analysis

- Savitzky-Golay filter was applied with
  - 46 frames (~1.5 s)
  - polynomial order 3
- Velocity vs. time computed from taking the derivative of Position vs. time
- Negative velocity minima is selected.



Results

Savitzky-Golay filter



By MothNik - Own work, CC0, https://commons.wikimedia.org/w/index.php?curid=164353357

Results

Data analysis

- Viscous drag force a trapped bead was calculated using Stokes' law
  - η : the dynamic viscosity of water at room temperature
  - r : bead radius
  - v : is the measured free-bead velocity at each flow rate
- The trap stiffness k was determined by fitting
  - a linear model to the measured force-displacement pairs across all flow rates:

$$F_{\mathrm{drag}} = 6\pi\eta r v$$

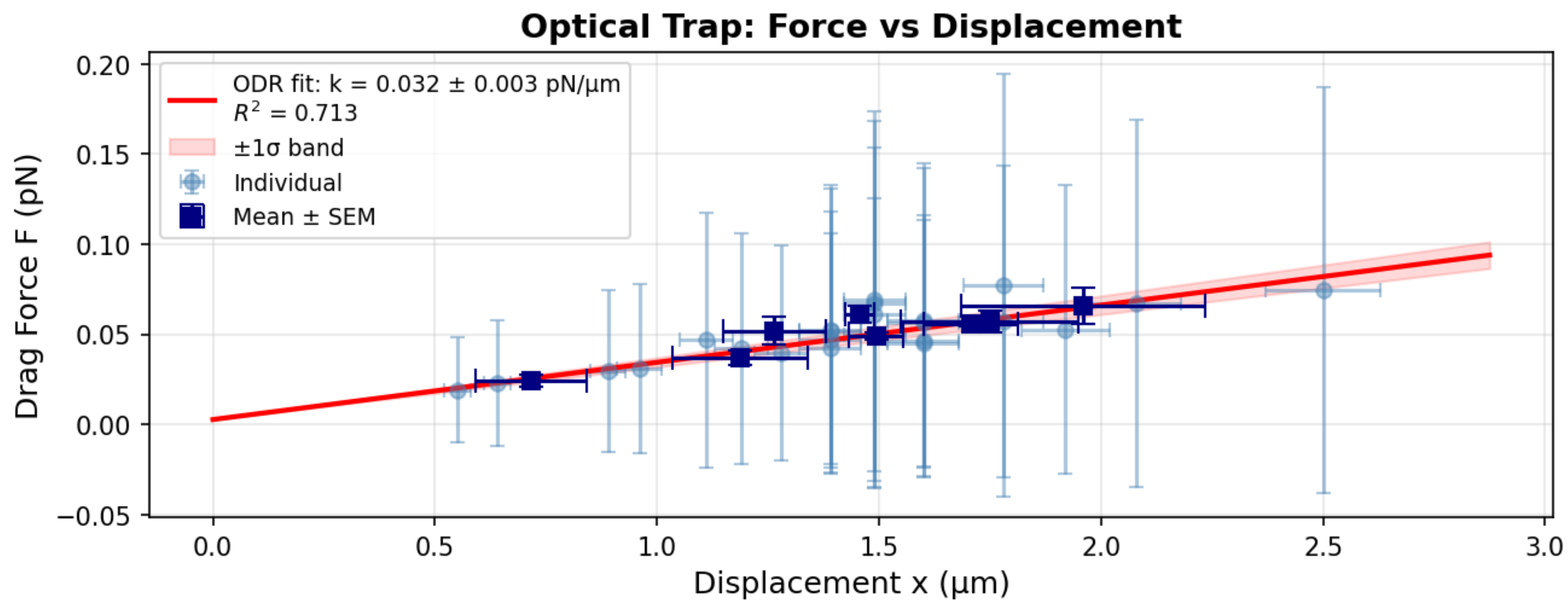
$$F = kx$$

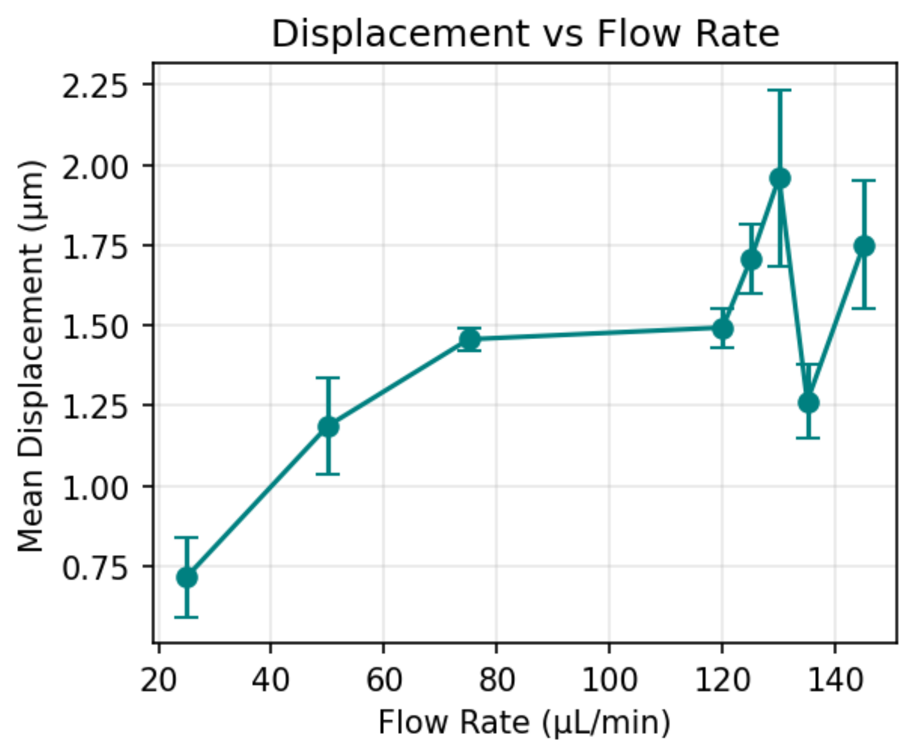
Displacement & Force vs. Flow Rate

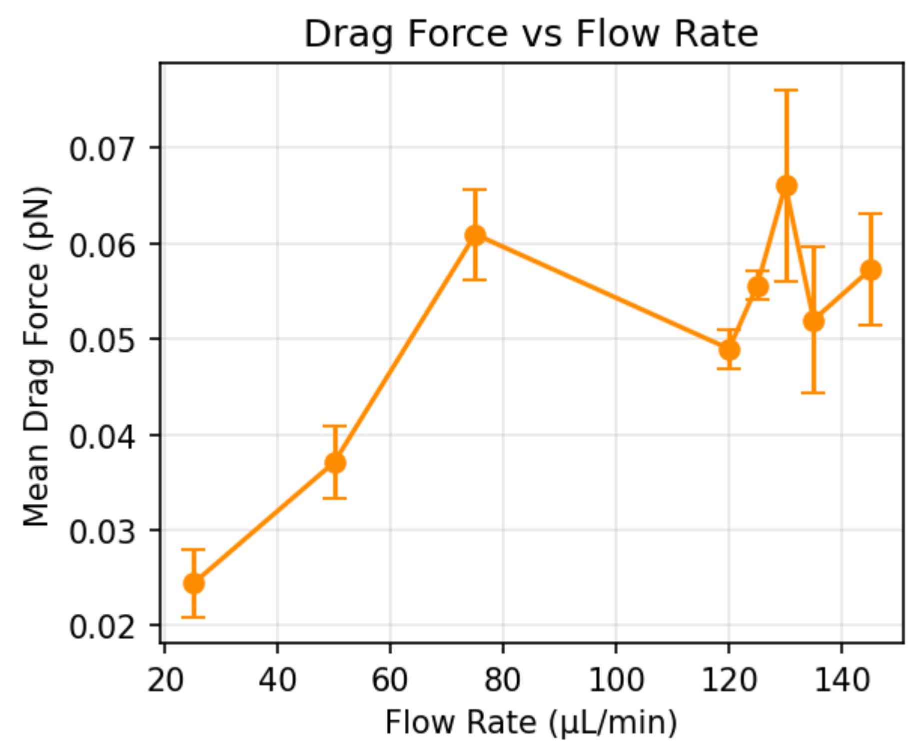
- (ODR) fit was applied to the force displacement data

- 1 sigma error band shown in red around ODR.

- Displacement is ~ 2 µm and drag force is ~ 0.065 pN at the max flow rate ~ 140 uL/min







Conclusion

- Successfully calibrated a single-beam optical trap using viscous drag from controlled flow rates
- Trap stiffness measured with the $R^2 = 0.713$; Confirmed Hookean behavior over 0 - 3 um displacement range
- Maximum sustainable trapping force ~ 0.065pN, consistent with the laser power and NA used.
- All data and analysis pipeline were all uploaded in the GitHub repository https://github.com/Lemon-Elf/PHYS5318.git; branch optical_tweezers